SEBASTIAN RAMIREZ GOMEZ

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OBJECTIVE

Dedicated university student soon to graduate in PHYSICAL ENGINEERING seeking a challenging role to leverage my strong problem-solving skills and interdisciplinary knowledge. With a solid foundation in PHYSICAL ENGINEERING, theoretical physics, and computational systems, I am eager to contribute to innovative projects and collaborate effectively within diverse teams.

OBJECTIVE

I am eager to deepen my understanding of MATERIALS SCIENCE AND ENGINEERING by pursuing a Master's degree in SOUTH KOREA. I believe this opportunity will not only enhance my technical skills but also allow me to contribute to the academic and research community through collaboration and knowledge sharing. By immersing myself in the SOUTH KOREA culture, I hope to cultivate valuable relationships and broaden my global perspective.

EDUCATION

Bachelor's degree in PHYSICAL ENGINEERING 2019 - Ongoing
Technological University of Pereira, Pereira, Colombia. CGPA : 4.63/5

Introduction to computational modeling methods in MATERIALS SCIENCE 2023 - 2024
National university of La Plata, La Plata, Argentina. CGPA : 9/10

Industrial technical bachelor, specialized in MECHATRONICS 2014 - 2019
Santa Rosa de Cabal Technological Institute, Santa Rosa de Cabal, Colombia, CGPA : 4.8/5

PROJECTS

DFT study of hydrogen adsorption on silicene and germanene, July 2023 - Ongoing
Major Project *Colombia*

- Solution to hydrogen storage problems.
- Computational modeling of two-dimensional structures in QE.
- Reduces the risk when transporting hydrogen.
- Complies with the hydrogen roadmap program.
- Promotes research methods in the country.

DFT study of germanene and silicene systems, February 2024 - Ongoing
Collaboration project *Mexico*

- Proposes new two-dimensional materials and studies their properties.
- Computational modeling of two-dimensional structures in VASP.
- Materials with potential application in renewable energy technologies and energy storage.
- Exchange of ideas between different research groups.

POSITION OF RESPONSIBILITY

Guide for New Students: GIMOSIC Group (May 2024 - Ongoing)
Mentored and supported new group members, facilitating their integration and success.

Group Leader: Computational Modeling and Simulation Research Group (November 2023 - Ongoing)
Overseen research projects, coordinated team efforts, and fostered a collaborative environment.

Student Representative: Hydrogen Storage Research (July 2023 - Ongoing)
Advocated for student interests, communicated research findings, and contributed to project planning.

SKILLS AND INTERESTS

TECHNICAL SKILLS

***Programming Languages*:**	Python, C++, MATLAB.
***Computational Tools*:**	Quantum Espresso (QE), Vienna Ab initio Simulation Package (VASP).
***Software Proficiency*:**	Microsoft Office Suite.

INTERDISCIPLINARY KNOWLEDGE

Materials Science:	Semiconductors, 2D materials, hydrogen, renewable energy
Computational Modeling:	Simulations of materials properties

SOFT SKILLS

Effective communication
Strong teamwork and collaboration
Problem-solving and critical thinking.

EXTRACURRICULAR ACTIVITIES AND ACHIEVEMENTS

Upcoming Conferences

- Presenter at the XII International Congress on Materials 2024 (Colombia)
- Participant in the World Conference on Applied Science, Engineering & Technology (Malaysia)

Recent Conferences

- Participant in the X Workshop on Novel Methods for Electronic Structure Calculations (Argentina, 2023)
- Participant in the Conference on Materials and Additive Engineering (Colombia, 2023)
- Presenter at the Statistics Conference of the UTP (Colombia, 2018)

Academic Excellence

- Achieved a 100 percentile in all of Colombia in the national 11th grade test (Colombia, 2019)
- First place in the third round of the national mathematics tests, saber 5 (Colombia, 2013)

Community Involvement

- Volunteering as a helper in communities in my region (Colombia, Ongoing)

Other Achievements

- Third place in the programming competition organized by INSILICO (Colombia, 2023)
- Member of the computational modeling and simulation research group (Colombia, Ongoing)
- Qualified for the national table tennis tournament (Colombia, 2016)